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AI Based Diabetes Prediction System

1. Project overview
   This AI-Based Diabetes Prediction System is designed to analyse patient health data and predict the likelihood of diabetes. This project will help healthcare professionals and individuals identify risk factors early, enabling timely preventive care and intervention.

   Objectives:
   - Develop a predictive model to classify whether a person is diabetic (1) or not (0).
   - Apply preprocessing, feature selection,and model evaluation on medical datasets.
   - Build an interactive interface to input patient details and get real time predictions.

2. Dataset Description
   This project utilized the Prima Indian Diabetes Dataset sourced from Kraggle.
   - Total records: 768 female patients of Prima Indian heritage, 21 years or older.
   - Input feature:
     1. Pregnancies: Plasma glucose concentration
     2. `Glucose`: Plasma glucose concentration.
     3. `Blood Pressure`: Diastolic blood pressure (mm Hg).
     4. `Skin Thickness`: Triceps skinfold thickness (mm).
     5. `Insulin`: 2-Hour serum insulin (mu U/ml).
     6. `BMI`: Body mass index (weight in kg/(height in m)^2).
     7. `Diabetes Pedigree Function`: A function calculating diabetes likelihood based on family history.
     8. Age: Age in years.

   - Target Variable : outcome (Class 0 = Non-Diabetic, Class 1 = Diabetic).

3. Data Preprocessing Techniques:
   To ensure the machine learning model received clean, mathematically sound data, the following steps were executed:-
   - Handling missing values: the missing values from the dataset were replaced with NaN and then imputed using the median value of their respective columns to prevent extreme outliers from skewing the data.
   - Feature scaling: As features like insulin have higher numerical ranges compared to the features like Diabetes Pedigree Function, Standard Scaler was applied. This standardised all numeric input feature to have a mean of 0 and a standard deviation of 1, ensuring no single feature dominated the model's decision making process.
   - Data splitting: the processed data was split into 80% training set and 20% testing set so the model could be evaluated on unseen data for a higher accuracy prediction.

4. Model training and Optimization strategy
   This project utilized Random Forrest Classifier as the core predictive algorithm.

   - Baseline model: an initial model was trained using 100 decision trees (n_estimators = 100). While overall accuracy was high, the recall score was slightly lower. In healthcare, we generally want recall to score higher to minimize False negative. This indicates a bias towards majority class (non-diabetic), meaning the dataset was missing too many actual diabetic cases.

   - Hyperparameter Tuning – Grid Search: to improve recall, GridSearchCV was utilized with 5-fold Cross Validation. The grid systematically tested various combinations of n_estimators, max_depth, min_samples_split, and min_samples_leaf. While the search identified the mathematically optimal parameters within the provided grid, however the cross validated recall score did not outperform or provided a higher range compared to baseline model. It could not overcome the dataset's structural bias.

   - Class Imbalance Resolution (SMOTE): Recognizing the underlying issue of class imbalance, the strategy shifted to data augmentation. The Synthetic Minority Over Sampling Technique was exclusively applied to the training data. SMOTE mathematically generated synthetic profiles for diabetic patients until the training set achieved a perfect balanced set (50/50).

- Final Model Training: Random Forest Model was trained on the newly balanced SMOTE dataset. This approach successfully removed the algorithmic bias, resulting in the model that was better at identifying true positive diabetic cases.

5. Final Evaluation Metrics:
   The final SMOTE-optimized Random Forest model was evaluated against the isolated 20% testing set. The model achieved the following matrices:

- Accuracy: 0.9481
- Precision: 0.9123
- Recall: 0.9455
- F1-Score: 0.9286
- ROC-AUC: 0.9831

6. Interactive Web application:

To make the model accessible, a front-end web application was built using **Streamlit**.

- Functionality: The app accepts manual inputs for the 8 medical features through user-friendly number fields.
- Integration: Behind the scenes, the app dynamically loads the saved `diabetes_scaler.pkl` to scale the user's input, and the `diabetes_model.pkl` to generate the prediction.
- Output: The interface immediately outputs a binary classification ("Diabetic" or "Non-Diabetic") alongside the exact risk probability percentage, providing actionable health analytics in real-time.